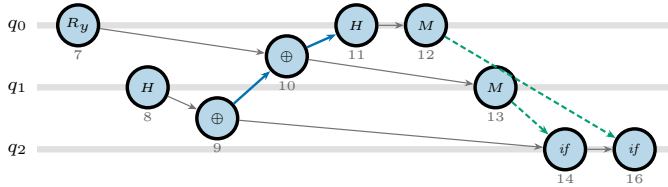
(a) Teleportation

```
 7 qc.ry(0.9, 0)                    # state to send
 8 qc.h(1)
 9 qc.cx(1, 2)                      # Bell pair q1, q2
10 qc.cx(0, 1)
11 qc.h(0)
12 qc.measure(0, 0)
13 qc.measure(1, 1)
14 with qc.if_test((c[1], 1)):
15     qc.x(2)
16 with qc.if_test((c[0], 1)):
17     qc.z(2)
```

(b) Wire-rule dependence graph, criterion q_2

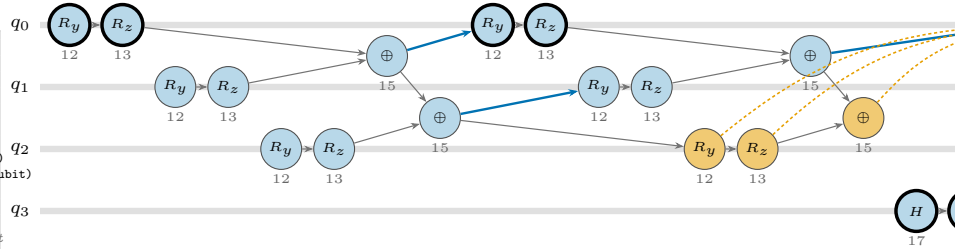


def-use: 4 ops, lines 8, 9, 14, 16, $F = 0.50$ classical PDG = wire: 9 ops, lines 7, 8, 9, 10, 11, 12, 13, 14, 16, $F = 1.00$

(c) VQE ansatz

```
 5 layers = 2
 6 rng = np.random.default_rng(7)
 7
 8 qc = QuantumCircuit(n + 1, n + 1)
 9
10 for layer in range(layers):
11     for qubit in range(n):
12         qc.ry(float(rng.uniform(0, np.pi)), qubit)
13         qc.rz(float(rng.uniform(0, 2 * np.pi)), qubit)
14     for qubit in range(n - 1):
15         qc.cx(qubit, qubit + 1)
16
17 qc.h(n)                           # auxiliary qubit
18 qc.measure(n, n)                  # mid-circuit measurement
19 with qc.if_test((qc.clbits[n], 1)):
20     qc.x(0)                      # flip conditioned on it
21 qc.measure(range(n), range(n))
```

(d) Wire-rule dependence graph, criterion q_0



classical PDG: 7 ops, lines 12, 13, 17, 18, 19, $F = 0.97$ QPDG: +3 non-dependences wire: 16 ops = dependences, $F = 1.00$

data control quantum, missing from the classical PDG entanglement closure (QPDG)
wire slice classical PDG slice added by QPDG, not a dependence small numbers: source line